

Software Design Pattern Course Suggestions

1. Importance of Design Pattern
2. Category of Design Pattern and Their Purpose
3. Difference Between Factory and Abstract Factory Pattern
4. Singleton Design Pattern
5. Case Study / Pattern Implementation
6. UML Class Diagram
7. Code Implementation
8. Code Smell Overview
9. SOLID Principles
10. Dependency Principle
11. Refactorization

Design Patterns to Study:

- Factory Pattern
- Singleton Design Pattern
- Abstract Factory Pattern
- Builder Pattern
- Adapter Pattern
- Decorator Pattern
- Chain of Responsibility Pattern

- Composite Pattern
- Proxy Pattern

Case Study: University HR System

- A university has several colleges. Each college has its own HR system and employees.

Topics:

1. Creating a single university instance
2. Creating employee objects based on designation
3. Converting different HR system formats into one HR system format
4. Teacher inherits from Employee class

Additional Topics:

- Single Tone (Singleton) Pattern
- Class Diagram
- Design Pattern Structure
- Purpose of Each Pattern
- Advantages and Usage of Design Patterns